

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score

# -----
# Load Dataset
# -----
df = pd.read_csv('/content/House Price Prediction Dataset.csv')

# -----
# Features & Target
# -----
X = df[['Area']]
y = df['Price']

# -----
# Train-Test Split
# -----
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# -----
# Model Training
# -----
model = LinearRegression()
model.fit(X_train, y_train)

# -----
# Prediction
# -----
y_pred = model.predict(X_test)

# Accuracy
print("R2 Score:", r2_score(y_test, y_pred))

# Predict custom value
predicted_price = model.predict(pd.DataFrame([[1100]], columns=['Area']))
print("Predicted Price for 1100 sq ft:", predicted_price[0])

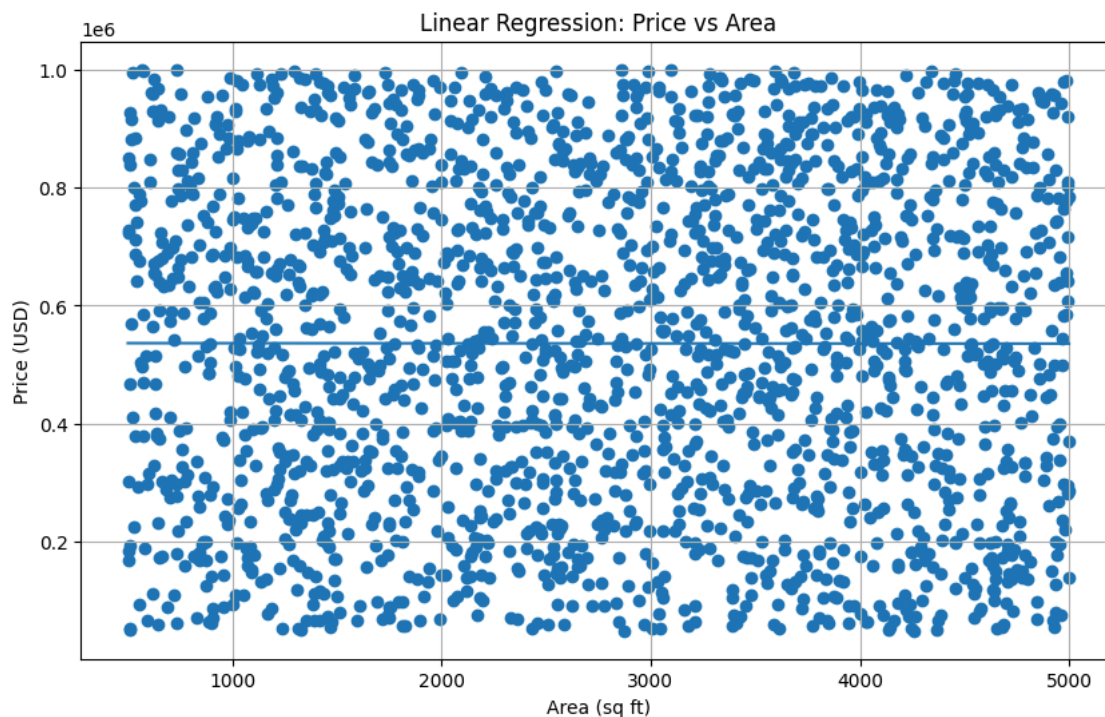
# -----
# Visualization (IMPORTANT FIX)
# -----
# Sort values for smooth line
sorted_idx = X['Area'].argsort()
X_sorted = X.iloc[sorted_idx]
y_sorted_pred = model.predict(X_sorted)

plt.figure(figsize=(10, 6))
plt.scatter(X, y)          # actual data points
plt.plot(X_sorted, y_sorted_pred)  # regression line

plt.xlabel("Area (sq ft)")
plt.ylabel("Price (USD)")
plt.title("Linear Regression: Price vs Area")
plt.grid(True)
plt.show()

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R2 Score: -0.0007365003351658128
 Predicted Price for 1100 sq ft: 536491.4322478768



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import StandardScaler

# -----
# Load Dataset
# -----
df = pd.read_csv('/content/House Price Prediction Dataset.csv')

# -----
# Create Binary Target
# -----
median_price = df['Price'].median()
df['Price_Above_Median'] = (df['Price'] > median_price).astype(int)

# -----
# Features & Target
# -----
X = df[['Area']]
y = df['Price_Above_Median']

# -----
# Train-Test Split
# -----
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# -----
# Feature Scaling (IMPORTANT)
# -----
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# -----
# Model Training
# -----
model = LogisticRegression()
model.fit(X_train_scaled, y_train)

# -----
# Accuracy
# -----
y_pred = model.predict(X_test_scaled)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```

# -----
# Prediction for 1100 sq ft
# -----
sample = scaler.transform(pd.DataFrame([[1100]], columns=['Area']))
prediction = model.predict(sample)
print("Will price be above median (0=No, 1=Yes)?", prediction[0])

# -----
# Visualization
# -----
x_values = np.linspace(X['Area'].min(), X['Area'].max(), 200)
x_scaled = scaler.transform(x_values.reshape(-1,1))
y_prob = model.predict_proba(x_scaled)[:,-1]

plt.figure(figsize=(10, 6))

# Actual data
plt.scatter(X, y, alpha=0.6)

# Sigmoid curve
plt.plot(x_values, y_prob, linewidth=2)

# Decision boundary (0.5 threshold)
plt.axhline(0.5, linestyle='--')
decision_boundary = x_values[np.argmin(np.abs(y_prob - 0.5))]
plt.axvline(decision_boundary, linestyle='--')

plt.xlabel("Area (sq ft)")
plt.ylabel("Probability (Above Median Price)")
plt.title("Logistic Regression - House Price Classification")
plt.grid(True)

```

Accuracy: 0.475

Will price be above median (0=No, 1=Yes)? 0

/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature name:
warnings.warn(

